

Data Mining Complete Exam Notes

CT 72502 (Elective I) · BEI / BEX IV-I · TU, IOE · tiers over 23 papers, 2069–2082 BS

5. Cluster Analysis

9 hrs · in 23/23 papers · 5 topics, 3 TOP tier · **366 marks, 18.9% of the archive** · **135 numerical (37%) + 231 theory** · numerals → Ch 5 companion.

Weight of the chapter:

- Basics & types of clustering **TOP 10/23** · K-means **TOP 8/23** · DBSCAN **TOP 8/23**
- Hierarchical **HOT 4/23** · Evaluation / issues **HOT 4/23**
- **K-means numerical appears in 8 papers, hierarchical dendrogram in 4.**

5.1 Basics and Algorithms

What is cluster analysis? Applications? How does it differ from classification? **8 5 2+6 3+4**

TOP 13/23 81 Ba, 80 Bh, 79 Bh, 75 Ch, 74 Ash, 73 Ch, 73 Shr, 72 Ch, 72 Ka, 71 Ch, 71 Shr, 70 Asa, 69 Ch

Definition

- **Cluster** = a collection of objects **similar within** the group and **dissimilar to** objects in other groups.
- **Cluster analysis** = grouping objects so that intra-cluster similarity is **maximised** and inter-cluster similarity **minimised**.
- **Unsupervised**: no class labels are given. The classes are the *output*, not the input.
- Also called **unsupervised classification** or segmentation.

Clustering vs classification 72 Ka **5**

	Classification	Clustering
Learning	Supervised	Unsupervised
Labels	Given in training	None; discovered
Goal	Predict a known label	Find the groups themselves
Evaluation	Accuracy vs truth	Cohesion / separation
Classes	Fixed, known	Number often unknown

Applications asked in 5 papers

- **Market segmentation**: group customers by behaviour.
- **Document / search-result grouping**, topic discovery.
- **Image segmentation** and pattern recognition.
- **Biology**: taxonomy of species, gene expression grouping.
- **Land use** from satellite data; city planning; earthquake epicentre grouping.
- **Outlier detection** — points in no cluster → Ch 6.
- **Data reduction / preprocessing** step for other algorithms.

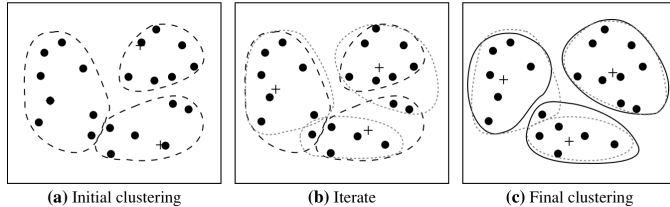
Requirements of a good clustering method

- Scalability; handles **mixed attribute types**; finds **arbitrary shapes**.
- Minimal domain knowledge for parameters; copes with **noise and outliers**.
- **Insensitive to input order**; handles high dimensionality; interpretable and usable.

Types of clusters (81 Ba asks this directly)

- **Well-separated**: every point is closer to all points in its own cluster than to any point outside. The ideal case.
- **Prototype-based (centre-based)**: each point is closer to its cluster's **centroid** (mean) or **medoid** (most central actual point) than to any other. → what K-means finds.
- **Contiguity-based (nearest neighbour)** 71 Shr 10: each point is closer to at least **one point** in its cluster than to any point outside. Handles chains and irregular shapes; **breaks down when noise bridges 2 clusters**. → single-link hierarchical, or DBSCAN.

Prototype-based, found by K-means iterating (§5.2)



(a) initial clustering around the seeds; (b) centroids recomputed and points reassigned; (c) final, once the assignment stops changing. H&K Fig 10.3.

- **Density-based**: a cluster is a **dense region** separated by low-density regions. Handles arbitrary shapes and noise. → DBSCAN.
- **Conceptual (shared-property)**: points share some general property, not just proximity. Needs a concept definition.

Types of clustering methods

- **Partitioning**: K-means, K-medoids/PAM, CLARA.
- **Hierarchical**: AGNES, DIANA, BIRCH, Chameleon.
- **Density-based**: DBSCAN, OPTICS, DENCLUE.
- **Grid-based**: STING, CLIQUE, WaveCluster.
- **Model-based**: EM, COBWEB, SOM.

Grid-based, in 3 steps (NEW)

- **Partition** the data space into a multi-dim grid of finite cells.
- **Compute cell statistics**: count, mean, density per cell.
- **Merge** adjacent cells above a density threshold into clusters.
- **Why it is the fast one**: cost depends on the **number of cells**, not the number of points → scales to huge datasets. Price paid: resolution is capped by cell size.

5.2 K-means Clustering

Write the K-means algorithm. Explain with example and state its limitations. 2+6 2+8 4+6 4+4

TOP 13/23 82 Bh, 81 Bh, 81 Ba, 80 Ba, 79 Ba, 78 Bh, 76 Ch, 75 Ash, 74 Ch, 74 Ash, 73 Ch, 71 Shr, 69 Ch Numerical in 7 of those → companion.

Algorithm (write these 5 steps, then compute)

1. Choose K , the number of clusters.
2. **Initialise** K centroids (usually K randomly chosen data points).
3. **Assign** each point to the cluster whose centroid is nearest (Euclidean).
4. **Recompute** each centroid as the **mean** of the points assigned to it.
5. **Repeat** 3–4 until the assignment stops changing (or centroids move less than a tolerance, or a max iteration count is hit).

Objective function

$$SSE = \sum_{i=1}^K \sum_{x \in C_i} dist(x, c_i)^2$$

- K-means **minimises SSE** (within-cluster sum of squared error).
- The mean is provably the point minimising SSE for a cluster — **that is why step 4 uses the mean**.
- Complexity $O(nKI d)$: n points, K clusters, I iterations, d dimensions. Linear in n → scales well.

Choosing K 80 Ba 4

- **Elbow method**: plot SSE against K ; take the K where the curve bends. SSE always falls as K rises, so the **bend**, not the minimum, is the answer.
- **Silhouette coefficient**: pick the K maximising it.
- **Domain knowledge**; cross-validation; gap statistic.
- Rule of thumb $K \approx \sqrt{n/2}$.

Limitations (asked with almost every numerical)

- **K must be given in advance**.
- **Sensitive to the initial centroids** → different seeds give different clusterings; converges to a **local** optimum, not the global one. *Fix*: run several times and keep the lowest SSE, or use K-means++.
- **Sensitive to outliers**: the mean is dragged by extreme values. *Fix*: K-medoids.
- Only finds **convex, roughly spherical, similar-sized** clusters. Fails on elongated or nested shapes.
- Cannot handle **categorical** data (no mean). *Fix*: K-modes.
- Empty clusters can occur; needs a re-seeding rule.

K-medoids (PAM)

- Uses an **actual data point** (the medoid) as the cluster centre instead of the mean.
- Minimises the sum of *absolute* distances rather than squared distances.
- **Far more robust to outliers** and works with any distance measure.
- Cost: $O(K(n - K)^2)$ per iteration $\rightarrow$ much slower; CLARA samples to scale it.

Bisecting K-means 71 Ch 10

Asked w/ "why is clustering unsupervised": no class labels, see §5.1.

- Variant of K-means giving a **partitional** or a **hierarchical** (divisive) clustering.
- = K-means + divisive hierarchical: K-means w/ $K = 2$ is the splitting engine.
- **Algorithm:**
 1. Init list of clusters = {one cluster holding all points}.
 2. **Pick** a cluster from the list to split:
 - largest cluster, or
 - cluster w/ highest SSE (least cohesive).
 3. **Bisect:** run basic K-means ($K = 2$) on it $\rightarrow$ 2 sub-clusters.
 4. Repeat step 3 for **ITER** trials (different initial centroids); keep the split w/ **lowest total SSE**.
 5. Replace parent by its 2 children in the list.
 6. Repeat 2-5 until list holds K clusters.

- How the hierarchy forms:

- Each bisection = 1 node split into 2 children.
- Recording the splits gives a top-down tree = **dendrogram**.
- Run to $K = n \rightarrow$ full hierarchy, 1 point per leaf.

- Advantages:

- **Less sensitive to initialisation** than plain K-means (only 2 centroids per trial, best of ITER).
- Final clusters can be refined by plain K-means using their centroids as seeds.
- Cheaper than agglomerative: $O(n)$ per split vs $O(n^2)$ distance matrix.

- Limitations:

- Split cannot be undone.
- No single global objective minimised (each split is locally optimal).
- Inherits K-means problems: noise, outliers, non-convex shapes, unequal sizes.

5.3 Hierarchical Clustering

What is hierarchical clustering? Agglomerative vs divisive. Draw the dendrogram. 2+7 8 4+4 10

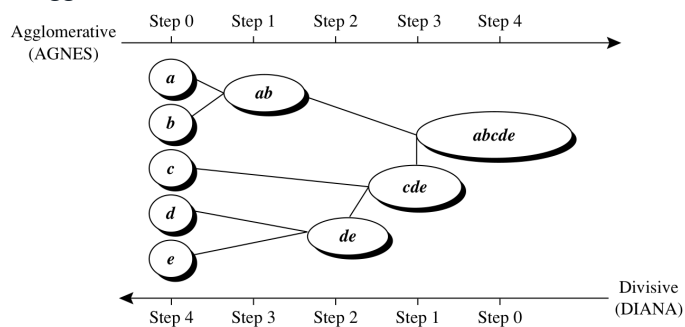
HOT 5/23 80 Bh, 80 Ba, 79 Ba, 76 Ch, 70 Ch 4 of the 5 are numericals → companion.

Idea

- Produces a **nested tree of clusters**, not a single flat partition.
- **No need to specify K in advance** — cut the dendrogram at any height to get any K .
- **Dendrogram**: tree diagram where the **height of a merge** = **the distance between the two clusters merged**.

Two directions

- **Agglomerative (AGNES)** — **bottom-up**, the common one.
 1. Start: every point is its own cluster.
 2. Merge the **two closest** clusters.
 3. Update the distance matrix.
 4. Repeat until one cluster remains.
- **Divisive (DIANA)** — **top-down**.
 1. Start: all points in one cluster.
 2. Split the cluster that is least cohesive.
 3. Repeat until every point is alone (or a stopping rule fires).
- **Divisive is more expensive** ($2^{n-1} - 1$ ways to split), so agglomerative dominates.



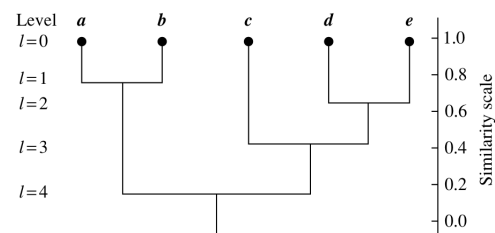
AGNES merges $a \dots e$ upward in steps 0 to 4; DIANA splits them downward in the reverse order. H&K Fig 10.6.

Linkage: how to measure cluster-to-cluster distance (this is the whole exam question)

- **Single link (MIN)**: distance = **closest pair** between the 2 clusters.
 - + Finds **arbitrary, elongated** shapes.
 - - **Chaining effect**: a line of noise points can bridge 2 real clusters.
- **Complete link (MAX)**: distance = **farthest pair**.
 - + Less sensitive to noise, gives compact balanced clusters.
 - - Biased to globular shapes; **breaks large clusters**.
- **Average link**: mean of all pairwise distances. A compromise.
- **Centroid / Ward's**: distance between centroids / merge that increases SSE least.

Pros and cons

- + No K needed; the dendrogram is highly interpretable; any K obtainable afterwards.
- - $O(n^2)$ **space** and $O(n^3)$ **time** → does not scale.
- - **Merges are irreversible** — a bad merge can never be undone (greedy).



The same merges as a dendrogram, level $l = 0$ (all apart) to $l = 4$ (one cluster); cut at any level for that many clusters. H&K Fig 10.7.

5.4 DBSCAN

What are core, border and noise points? Write the DBSCAN algorithm and how it handles noise. 3+5

8 8+2 7

TOP 12/23 82 Bh, 81 Bh, 79 Bh, 79 Ba, 78 Bh, 76 Ash, 75 Ch, 75 Ash, 74 Ash, 72 Ka, 70 Ch, 70 Asa

Two parameters

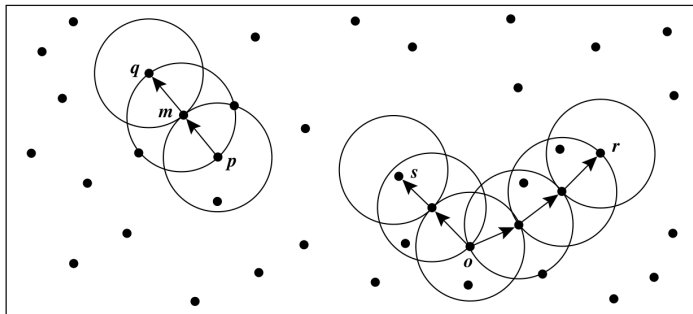
- ϵ (Eps) = the neighbourhood radius.
- **MinPts** = minimum number of points required inside that radius.
- ϵ -neighbourhood: $N_\epsilon(p) = \{q \in D \mid \text{dist}(p, q) \leq \epsilon\}$

Three kinds of point (learn these exactly)

- **Core point**: $|N_\epsilon(p)| \geq \text{MinPts}$. Sits in the *interior* of a cluster.
- **Border point**: fewer than MinPts neighbours, **but lies within ϵ of a core point**. On the *edge*.
- **Noise point**: neither. **Discarded – this is how DBSCAN handles noise.**

The reachability definitions 74 Ash short note

- **Directly density-reachable**: q is directly density-reachable from p if p is a **core point** and $q \in N_\epsilon(p)$.
- **Density-reachable**: there is a **chain** $p_1, \dots, p_n$ where each p_{i+1} is directly density-reachable from p_i . **Not symmetric** (the end may be a border point).
- **Density-connected**: p and q are density-connected if some object o exists from which **both** are density-reachable. **This one IS symmetric.**
- **A cluster** = a maximal set of density-connected points.



Circles are ϵ -neighbourhoods, MinPts = 3. Left: m, p are core, q is density-reachable from p through m , but not the reverse (q is a border point). Right: r and s are density-connected through o . H&K Fig 10.14.

Algorithm

1. Label every point as core, border or noise by counting its ϵ -neighbours.
2. Discard the noise points.
3. Put an edge between every pair of core points within ϵ of each other.
4. Each **connected group of core points** becomes a cluster.
5. Assign each border point to a cluster of one of its associated core points.

Choosing the parameters 78 Bh asks this

- **k -distance plot**: for each point compute the distance to its k th nearest neighbour ($k = \text{MinPts}$), sort descending, plot. The **sharp bend (knee)** gives ϵ .
- Rule of thumb: $\text{MinPts} \geq d + 1$, commonly $2 \times d$ for d dimensions; $\text{MinPts} = 4$ for 2-D.

Pros

- Finds **arbitrarily shaped clusters**. • **K is not needed.**
- **Explicitly identifies noise / outliers** rather than forcing every point into a cluster.
- Insensitive to the order of points (except some border cases).

Cons and their fixes 76 Ash asks “how do we avoid the issues”

- **Fails on clusters of varying density** – one $(\epsilon, \text{MinPts})$ pair cannot suit both a dense and a sparse cluster. **Fix: OPTICS**, which orders points across a range of ϵ .
- Very sensitive to the 2 parameters. **Fix: the k -distance plot.**
- Degrades in **high dimensions** (distance loses meaning). **Fix: reduce dimensions first.**
- A border point reachable from 2 clusters is assigned by processing order.

5.5 Issues: Evaluation, Scalability, Comparison

Explain the different measures of cluster validity. How do you evaluate the clusters generated? 10 4

3

HOT 6/23 80 Bh, 74 Ch, 72 Ka, 71 Ch, 70 Asa, 69 Ch

Why it is hard

- Clustering is unsupervised, so **there is no ground truth to compare against**.
- Any algorithm will return clusters even on **random data** — so the first question is whether cluster structure exists at all (**cluster tendency**).

Three families of measure

- **Internal (unsupervised)** — uses only the data.
 - **Cohesion** (compactness): within-cluster SSE. Lower is better.
 - **Separation**: between-cluster sum of squares. Higher is better.
 - **Silhouette coefficient** for point i :

$$s = \frac{b - a}{\max(a, b)}$$

a = mean distance to points in its own cluster, b = mean distance to the nearest other cluster. Range $[-1, 1]$; near **1** good, near **0** on a boundary, **negative** = **probably in the wrong cluster**.

- **External (supervised)** — compares against known labels.
 - **Purity, entropy, precision/recall/F, Rand index, Jaccard coefficient**.
- **Relative** — compares 2 clusterings, eg by SSE or silhouette, to choose K .

Other issues

- **Determining K** — see the elbow and silhouette methods in §5.2.
- **Scalability**: hierarchical is $O(n^3)$; K-means is linear but needs many passes; BIRCH and CLARA exist to scale.
- **High dimensionality**: distances converge → use subspace clustering (CLIQUE) or reduce dimensions first (PCA, Ch 2).
- **Mixed attribute types**: needs a mixed dissimilarity measure (Ch 2, §2.4).
- **Outliers** distort centroids → K-medoids or DBSCAN.
- **Interpretability**: a cluster is only useful if it can be described.

Choosing an algorithm 79 Bh 2+6

Internet marketer: segment users by top search keywords, top URLs, recent purchases, usage level, peak hour, peak day.

- **Read the data first**:
 - keywords, URLs, vendor, product, day = **nominal**
 - qty, amount, hour = **numeric**; usage level = **ordinal**
 - no labels, K unknown
- **Algorithm 2: need a method that handles mixed data**.
 - **K-prototypes**: K-means for numeric attrs + K-modes for nominal. Scales to millions of users.
 - Or **hierarchical / DBSCAN** on a **mixed dissimilarity** (Ch 2, §2.4) → no K needed, dendrogram shows natural segment count.
 - Plain K-means unsuitable: mean of keywords/URLs undefined.
- **Validation 6**:
 1. **Cluster tendency**: check structure exists at all (Hopkins statistic) before trusting any segment.
 2. **Choose K** : run $K = 2$ to 10; elbow in SSE, peak of avg silhouette.
 3. **Internal**: cohesion (within-cluster SSE, low), separation (between, high), silhouette per segment → negative values flag misplaced users.
 4. **Stability**: re-cluster on random subsamples / different seeds; same segments reappear → real.
 5. **External**: if some users carry known labels (eg loyal vs churned), purity, entropy, Rand index.
 6. **Business validation**: profile each segment (dominant keywords, spend, peak time); segment must be describable + actionable, eg “weekend night gamers, high spend”.
 7. **Test**: run a targeted campaign per segment; response rate difference confirms usefulness.

5.6 PYQ Mapping

N = solved in the Ch 5 Numerical companion.

5.1 Basics

1. **TOP 10/23** Cluster analysis + applications + types of clusters. 8 81 Ba 5 72 Ka 2+6 75 Ch 8 73 Shr 3+4 80 Bh
2. **PIN 1/23** Why is clustering unsupervised? + hierarchical clusters via **Bisecting K-means**. 10 71 Ch
3. **PIN 2/23** Types of clustering methods + DBSCAN w/ example 10 72 Ch; partition methods + hierarchical w/ example 10 74 Ash
4. **PIN 1/23** What is a **contiguous cluster**? Explain an algorithm that generates them. 10 71 Shr
5. **PIN 1/23** Internet-marketer segmentation: which algorithm, and how to validate. 2+6 79 Bh

5.2 K-means

6. **PIN 1/23** K-means algorithm + strategy for choosing optimal K . 4+4 80 Ba
7. **N** Cluster for $K = 2$ w/ Euclidean distance + demerits. 5+3 81 Ba → **the paper prints NO data table; see the companion.**
8. **N** Two clusters from the 6-instance dataset. 2+6 78 Bh 2+8 74 Ash
9. **N** K-means w/ limitations, 6-row dataset. 4+6 75 Ash
10. **N** K-means w/ limitations, 7-row dataset. 10 71 Shr
11. **N** Matrix X , $K = 3$, given initial centres; final centres + iteration count. 8 76 Ch
12. **PIN 1/23** Hierarchical vs partitioning + how K-means is applied, w/ example. 2+8 74 Ch

5.3 Hierarchical

13. **N** Draw the dendrogram for the 6×6 distance matrix. 2+7 80 Bh
14. **N** **Complete-linkage** clustering + dendrogram from coordinates. 8 80 Ba
15. **N** **Single-link and complete-link** dendrograms for 6 objects. 4+4 76 Ch
16. **PIN 1/23** Agglomerative vs divisive w/ illustrative examples. 10 70 Ch

5.4 DBSCAN

17. **TOP 4/23** Core / border / noise points + DBSCAN algorithm + noise handling. 3+5 79 Bh 8 75 Ch 8+2 76 Ash 7 72 Ka
18. **PIN 1/23** Neighbourhood quantification, how clusters are found, choosing Eps and MinPts, and the $\varepsilon_1 < \varepsilon_2$ subset proof. 2+2+2 78 Bh
19. **PIN 2/23** Density-based cluster + an algorithm that generates them. 8 70 Ch 8 75 Ash
20. **PIN 1/23** Short note: Density reachable and Density connected. 4 74 Ash

5.5 Issues

21. **PIN 1/23** Different measures of cluster validity. 10 71 Ch
22. **PIN 3/23** How do you evaluate the clusters generated? 4 80 Bh; short notes: Issues in clustering 74 Ch, Cluster evaluation 72 Ka. 3

Papers added by the 2026-09 merge

The six papers of DM4.1BCT.pdf (2082 Bh, 2081 Bh, 2079 Ba, 2073 Ch, 2070 Asa, 2069 Ch), indexed here for the first time — the notes were built before that scan was merged. N = solved in this chapter's companion. **verified/added**

- **N** Core, border and noise points + **DBSCAN on 8 points**, $\varepsilon = 2$, MinPts = 2. 3+5 82 Bh **verified/added**
- **N** K-means, 2 clusters from a 6-instance dataset. 2+8 82 Bh
- **N** K-means from **a single given centroid** $(2, 3)$, $k = 2$, + problems and fixes. 5+3 81 Bh
- **N** K-means with the **Manhattan** distance, 3 centres, **stop after 3 iterations**. 8 79 Ba **verified/added**
- **N** Clustering vs classification + 1-D K-means on 7 points + **compute SSE**. 4+8 73 Ch **verified/added**
- Hierarchical clustering w/ a dendrogram example. 6 79 Ba
- What are density-based clusters? + DBSCAN w/ example. 10 70 Asa
- Measures that can be used to **compare two clusters** 70 Asa · issues in cluster validation 69 Ch. 6 6
- K-means w/ examples 69 Ch · short notes **DBSCAN clustering** 81 Bh, 79 Ba. 10 4 4

Coverage check

- All Ch 5 PYQs covered. **11 are numerical** → companion.
- **Known defect in the source** 81 Ba: Q4b says “cluster the following given data for $K = 2$ ” but **the paper prints no data table** — it jumps straight to Q5. Confirmed against the scan. The companion supplies a substitute dataset. **verified/added**

- **Gaps filled by research** verified/added : the **silhouette coefficient** formula, the **k -distance plot** for choosing ϵ , and the formal **density-reachable vs density-connected** distinction (your notes use the terms without defining them).

Sources used for this chapter

- **Han & Kamber 3rd ed** ch 10 (basic cluster analysis, partitioning, hierarchical, density-based, evaluation) and ch 11 (advanced). **Primary authority.**
- **DBK Lectures 15–16**, data_mining_/Chapter 5- Clustering.pdf, Clustering.pdf, ch7_ClusterAnalysis.ppt, Chapter7_3.ppt, kmedoids.doc.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
DM	data mining	DW	data warehouse
DB	database	DBMS	database management system
KDD	knowledge discovery in databases	DSS	decision support system
OLTP	on-line transaction processing	OLAP	on-line analytical processing
ROLAP	relational OLAP	MOLAP	multidimensional OLAP
HOLAP	hybrid OLAP	ETL	extract, transform, load
BI	business intelligence	ER	entity-relationship
attr(s)	attribute(s)	dim(s)	dimension(s)
multidim	multidimensional	txn(s)	transaction(s)
info	information	ops	operations
repo	repository	freq	frequent / frequency
config	configuration	no.	number
PCA	principal component analysis	SVD	singular value decomposition
SMC	simple matching coefficient	KNN	k -nearest neighbour
ANN	artificial neural network	POS	point of sale
FK	foreign key	UI	user interface
GIGO	garbage in, garbage out	w/, w/o	with, without
rm	remove	vs	versus
Eg	example / for example	+Fig	draw a figure
→	leads to / therefore	+	and

Exam-paper month codes (used in the year tags): Ba = Baishakh · Jth = Jestha · Asa = Ashar · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Mng = Mangsir · Po = Poush · Ma = Magh · Ch = Chaitra.

Tier chips: TOP /23 = topic asked in 7 or more of the 17 papers · HOT /23 = 4–6 papers · PIN /23 = 1–3 papers but worth ≥ 8 marks. Mark chips such as 2+5 are the marks that question actually carried. verified/added marks anything researched and added that was not in the source notes.